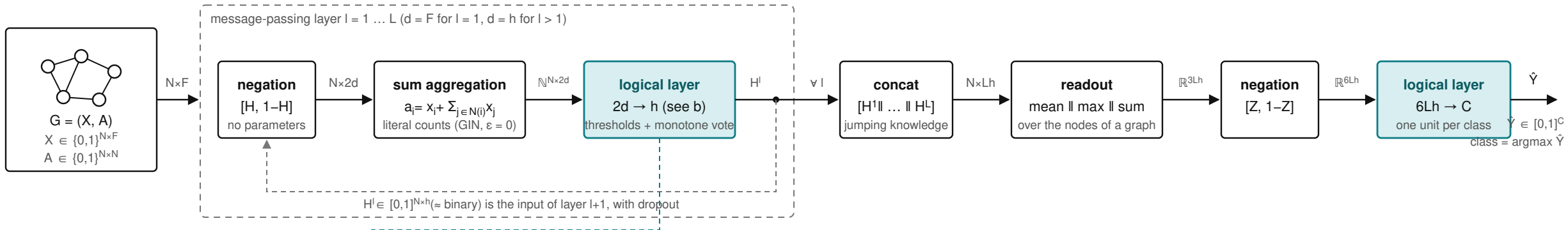
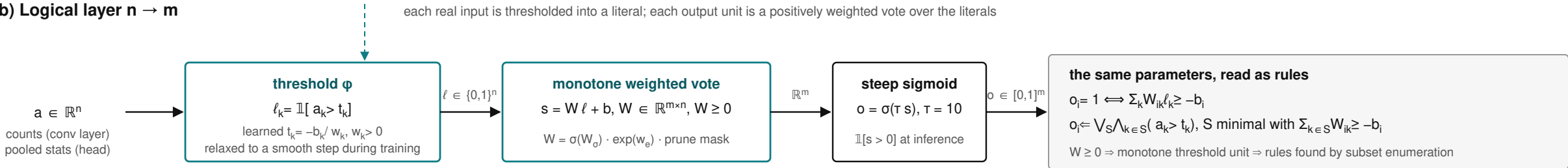


(a) LogiX-GIN forward pass



(b) Logical layer $n \rightarrow m$



Training: each logical layer first learns to reproduce the binary output of the matching layer of a Gumbel-binarised GIN teacher (layer-wise distillation), then the whole model is fine-tuned end-to-end on the labels. Regularisers: L1 on the gates $\sigma(W_o)$ (short rules), entropy on ℓ and o (binary activations).